```
(%i6) string(scsimp(expr, eq1, eq2));
(%o6)                             c^2-8
(%i7) quit();
```

Unlike Octave, Maxima, by default, doesn't evaluate its expressions; it only simplifies them. This means that expressions with integers like *cos(1)*, *exp(2)*, *sqrt(3)*, etc, may remain as they are in the most simplified form, without being evaluated to their respective float numerical values. In such cases, we may force evaluation by passing the option numer. Similar evaluation can be achieved for predicates, using *pred*:

```
$ maxima -q
(%i1) cos(1);
(%o1)                            cos(1)
(%i2) cos(1), numer;
(%o2)                       .5403023058681398
(%i3) sqrt(7);
(%o3)                            sqrt(7)
(%i4) sqrt(7), numer;
(%o4)                       2.645751311064591
(%i4) 1 + 2 > 9;
(%o4)                             3 > 9
(%i5) 1 + 2 > 9, pred;
(%o5)                             false
(%i6) string(%e^%pi < %pi^%e); /* string to print it on one line
*/
(%o6)                        %e^%pi < %pi^%e
(%i7) %e^%pi < %pi^%e, pred;
(%o7)                             false
(%i8) quit();
```

Summation simplifications

Symbolical representation and manipulations of summations can be beautifully achieved by using *sum()* and the option *simpsum*. Check out the code execution below:

```
$ maxima -q
(%i1) sum(a * i, i, 1, 5);
(%o1)                             15 a
(%i2) sum(a, i, 1, 5);
(%o2)                              5 a
(%i3) sum(a * i, i, 1, n);
                                    n
                                   ====
                                   \
(%o3)                           a   >    i
                                   /
                                   ====
                                   i = 1
(%i4) simpsum: true$ /* Enable simplification of summations */
(%i5) string(sum(a * i, i, 1, n)); /*string to print on a line*/
(%o5)                          a*(n^2+n)/2
(%i6) string(sum(i^2 - i, i, 1, n) + sum(j, j, 1, n));
(%o6)                       (2*n^3+3*n^2+n)/6
(%i7) string(factor(sum(i^2 - i, i, 1, n) + sum(j, j, 1, n)));
(%o7)                       n*(n+1)*(2*n+1)/6
(%i8) quit();
```

END

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